NADI: Equipment Checklist

Huawei ICT Competition 2026–2027, Innovation Track · UTM · For Dr Asri

Tick	No	Item	Qty	What it does	Status	Link
☐	1	ESP32-S3 board (DIYMORE N16R8, soldered pins)	4	Wi-Fi sensor that detects people in a room	To buy (~RM31 each)	Shopee
☐	2	DHT22 temperature sensor	1	Checks if the aircon is on	To buy (~RM15)	Shopee
☐	3	KY-005 IR transmitter	2	Turns the aircon off, like a remote	To buy (~RM4 each)	Shopee
☐	4	KY-022 IR receiver	1	Copies the aircon remote's signal	To buy (~RM4)	Shopee
☐	5	Breadboard (830 holes)	2	Connects parts without soldering	To buy (~RM7 each)	Shopee
☐	6	Jumper wires (F-F + M-M, 20 cm)	2 sets	Connects sensors to the boards	To buy (~RM6 each)	Shopee
☐	7	USB-C cables	4	Programs and powers the boards	Own	-
☐	8	Phone chargers (5V)	2	Powers the sensors in the room	Own	-
☐	9	Power banks	2	Powers the sensors at the TRX demo	Own	-
☐	10	Huawei Cloud vouchers	1	Runs the AI, dashboard and alerts online	**Needs to be provided**	-
☐	11	Wi-Fi access point + controller, **any brand UTM has** (Huawei AirEngine, Cisco, Aruba, Ruckus, Ubiquiti UniFi, TP-Link Omada, etc.)	1 set	Optional: counts phones and laptops connected in each room. We need admin access to read the device count (SNMP or API).	**Needs to be provided** (loan)	-
☐	12	GL.iNet Opal router *(only if UTM has no AP for row 11)*	1	Same job as row 11	Optional (~RM170 to 190)	GL.iNet

Must-haves: rows 1 and 10. Rows 11 and 12 are optional; NADI works without them. Rows 1 to 6 cost about RM180 to 200 in total (rough Shopee prices).